

Digital Music Store Analysis Using SQL

1. Project Objective

The objective of this project is to analyze **digital music store sales and customer data** using SQL to understand customer purchasing behavior, sales performance, popular music genres, artists, albums, and geographic trends. The analysis aims to identify key revenue drivers, high-value customers, and opportunities for improving sales and customer engagement.

2. Dataset Description

The dataset contains information related to customers, invoices, invoice items, tracks, albums, artists, genres, and employees.

Entity Description

Customer : Customer information such as name, country and city

Invoice : Customer purchase/transaction information

Invoice_Line : Individual tracks purchased in each invoice

Track : Information about individual music tracks

Album : Album information

Artist : Artist information

Genre : Music genre information

Employee : Employee/support information

3. Important Business Questions Answered Through SQL

SQL is used to answer business questions directly from the relational database.

Q 1 -- Who is the senior most employee based on job title?

	employee_id	last_name	first_name	title
▶	1	Adams	Andrew	General Manager

Q 2 -- Which countries have the most invoices?

	billing_country	total_billing
▶	USA	131
	Canada	76
	Brazil	61
	France	50
	Germany	41
	Czech Republic	30
	Portugal	29
	United Kingdom	28
	India	21
	Ireland	13
	Chile	13
	Finland	11

Result 21 ×

Q 3 -- What are top 3 values of total invoices?

	total
▶	23.759999999999998
	19.8
	19.8

Q 4 -- Which city has best customers? We would like to throw a promotional music Festival in the city we made the most money. Write a query that returns one city that has the highest some of invoice totals. Return both the city name and some of all invoice totals.

	billing_city	total_billing
▶	Prague	273.24000000000007

Q 5 -- Who is the best customer? The customer who has spent the most money will be declared the best customer. Write a query that returns the person who has spent the most money.

	customer_id	first_name	last_name	total_money
▶	5	František	Wichterlová	144.54000000000002

Q 6 -- Write query to return the email, first name, last name and genre of all rock music listeners. Return your list alphabetically by email starting with A.

	first_name	last_name	email
▶	Aaron	Mitchell	aaronmitchell@yahoo.ca
	Alexandre	Rocha	alero@uol.com.br
	Astrid	Gruber	astrid.gruber@apple.at
	Björn	Hansen	bjorn.hansen@yahoo.no
	Camille	Bernard	camille.bernard@yahoo.fr
	Daan	Peeters	daan.peeters@apple.be
	Diego	Gutiérrez	diego.gutierrez@yahoo.ar
	Dan	Miller	dmiller@comcast.com
	Dominique	Lefebvre	dominiquelefebvre@gmail.com
	Edward	Francis	edfrancis@yahoo.ca
	Eduardo	Martins	eduardo@woodstock.com.br
	Ellie	Sullivan	ellie.sullivan@shaw.ca

Result 15 x

Q 7 -- Let's invite the artist who have written the most rock music in our dataset. Write a query that returns the artist name and total track count of the top 10 rock bands.

	artist_id	name	no_of_songs
▶	1	AC/DC	18
	3	Aerosmith	15
	8	Audioslave	14
	22	Led Zeppelin	14
	4	Alanis Morissette	13
	5	Alice In Chains	12
	23	Frank Zappa & Captain Beefheart	9
	2	Accept	4

Q 8 -- Return all the track names that have a song length longer than the average song length. Return the Name and Milliseconds for each track. Order by the song length with the longest song listed first.

	name	milliseconds
▶	How Many More Times	711836
	Advance Romance	677694
	Sleeping Village	644571
	You Shook Me(2)	619467
	Talkin' 'Bout Women Obviously	589531
	Stratus	582086
	No More Tears	555075
	The Alchemist	509413
	Wheels Of Confusion / The Straightener	494524
	Book Of Thel	494393
	You Oughta Know (Alternate)	491885
	Terra	482429

track 17 x

Q 9 -- Find how much amount spent by each customer on artists? Write a query to return customer name, artist name and total spent.

	customer_id	first_name	last_name	artist_name	total_spent
▶	54	Steve	Murray	AC/DC	17.82
	53	Phil	Hughes	AC/DC	10.89
	21	Kathy	Chase	AC/DC	10.89
	49	Stanisław	Wójcik	AC/DC	9.9
	1	Luís	Gonzálves	AC/DC	7.920000000000001
	24	Frank	Ralston	AC/DC	7.920000000000001
	31	Martha	Silk	AC/DC	3.96
	16	Frank	Harris	AC/DC	2.9699999999999998
	42	Wyatt	Girard	AC/DC	2.9699999999999998
	6	Helena	Holmér	AC/DC	2.9699999999999998
	38	Niklas	Schröder	AC/DC	2.9699999999999998
	35	Madalena	Sampaio	AC/DC	2.9699999999999998

Result 19 x

Q 10 -- We want to find out the most popular music Genre for each country. We determine the most popular genre as the genre with the highest amount of purchases. Write a query that returns each country along with the top Genre. For countries where the maximum number of purchases is shared return all Genres.

	country	genre_id	name	purchase	m
▶	Argentina	1	Rock	1	1
	Australia	1	Rock	18	1
	Austria	1	Rock	6	1
	Belgium	1	Rock	5	1
	Brazil	1	Rock	26	1
	Canada	1	Rock	57	1
	Chile	1	Rock	7	1
	Czech Republic	1	Rock	14	1
	Denmark	1	Rock	6	1
	Finland	1	Rock	6	1
	France	1	Rock	26	1
	Germany	1	Rock	28	1

Result 20 x

16. Project Outcome

The Digital Music Store Analysis project demonstrates how raw transactional data can be transformed into actionable business information.

17. Conclusion

The Digital Music Store Analysis project provides a structured approach to understanding the performance of a digital music business.